

Maximal Size of the Universe from ξ

Schwarzschild Radius, Hubble Scale and the Complete Scale Hierarchy

Johann Pascher · March 2026

<https://github.com/jpascher/T0-Time-Mass-Duality>

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Correction notes

Corrections from the calculation check (30 Sept. 2026). The following places are corrected or annotated in the text; each carries a dated note with the previous value:

- Planck length: $\ell_P = \sqrt{G} = \xi/(2\sqrt{m_e})$ (dimension $[E^{-1/2}]$) $\rightarrow \ell_P = \sqrt{\hbar G_{SI}/c^3}$ with $G_{SI} = \frac{\xi^2}{4m_e} C_{conv} K_{frak}$; likewise in the derivation chain (there previously also with C_{dim}).
- $E_0 = \sqrt{m_e m_\mu} = 7.398 \rightarrow \sqrt{m_e m_\mu / K_{frak}} = 7.398$ (bare 7.348, R72), two places.
- Scale ratio $R_H/L_0 = 6.49 \times 10^{64}$: 57 $\rightarrow$ 65 orders of magnitude; $z \rightarrow \infty$ beyond R_H annotated (the formula $z \approx \xi \ln(d/\ell_P)$ gives only 0.019 at R_H).
- (2 Oct. 2026) Notation f : structure factor $f(n, l, j)$ — not a frequency; note in the text (Doc. 190, R147).

Abstract

ξ is the single fundamental parameter of T0 theory. The complete chain reads: $\xi \rightarrow \{m_e, m_\mu, \dots\} \rightarrow G \rightarrow \ell_P$, with $L_0 = \xi \cdot \ell_P$ as sub-Planck scale of spacetime granulation. The maximal scale R_H is written in the form $E_H = E_0 \cdot \xi^{41/4}$; the exponent 41/4, however, is not derived from ξ but calibrated to the measured H_0 (Doc. 190, R35/R36). The ratio $R_H/L_0 = 6.49 \times 10^{64}$ is therefore the measured cosmic scale in ξ -notation, not an independent ξ result.

The question of a universal maximum size of the universe is, however, not meaningfully posed in FFGFT: **every physical scale possesses its own system-specific upper bound** $R_{torus}(m) = \hbar/mc$. There is no single cosmic upper bound valid for all systems — R_H is the upper bound of the cosmological sector, not of the universe as a whole.

1 Derivation Chain: From ξ to Both Scale Limits**The Single Fundamental Parameter**

T0 theory has one single fundamental parameter (Document 013):

$$\xi = \frac{4}{3} \times 10^{-4} \quad (1)$$

All physical quantities derive from it. The complete chain reads:

$$\xi \rightarrow \{m_e, m_\mu, m_\tau, \dots\} \rightarrow G \rightarrow \ell_P \rightarrow c \quad (2)$$

with $L_0 = \xi \cdot \ell_P$ as the fundamental sub-Planck scale of spacetime granulation.

Derivation of Particle Masses from ξ

The lepton masses follow from ξ via the geometric quantum number factor $f(n, l, j)$ and the T0 scaling factor $S_{T0} = 1 \text{ MeV}/c^2$:

$$m_\ell = \frac{f(n, l, j)^2}{\xi^2} \cdot S_{T0} \quad (3)$$

The geometric mean of the lepton masses yields the characteristic energy scale:

$$E_0 = \sqrt{\frac{m_e \cdot m_\mu}{K_{frak}}} = 7.398 \text{ MeV} \quad (4)$$

(Corrected on 30 Sept. 2026: previously $E_0 = \sqrt{m_e m_\mu} = 7.398 - \sqrt{0.511 \times 105.66} = 7.348 \text{ MeV}$; 7.398 is the fractally corrected value $7.348/\sqrt{K_{\text{frak}}}$ (R72). Likewise below.) from which the fine-structure constant follows without free parameters:

$$\alpha = \xi \cdot \frac{E_0^2}{(1 \text{ MeV})^2} = \frac{1}{137.036} \quad \checkmark \quad (5)$$

(Note of 2 Oct. 2026 on notation: here $f(n,l,j)$ is the geometric structure factor of the quantum numbers, not a frequency and not the base winding number $f = 1/\xi$. Cf. Doc. 190, R147.)

Gravitational Constant and Planck Length

The gravitational constant follows from ξ and m_e (Documents 012, 013):

$$G_{\text{SI}} = \frac{\xi^2}{4m_e} C_{\text{conv}} K_{\text{frak}} \quad \Rightarrow \quad \ell_{\text{P}} = \sqrt{\frac{\hbar G_{\text{SI}}}{c^3}} \quad (6)$$

With $G_{\text{SI}} = 6.6745 \times 10^{-11} \text{ m}^3 \text{kg}^{-1} \text{s}^{-2}$ this gives: (Corrected on 30 Sept. 2026: previously $\ell_{\text{P}} = \sqrt{G} = \xi/(2\sqrt{m_e}) \approx 9.33 \times 10^{-5}$ "after SI conversion" – $\xi^2/(4m_e)$ has dimension $[E^{-1}]$, its square root $[E^{-1/2}]$ is not a length; as in Doc. 013, ℓ_{P} follows via G_{SI} .)

$$\ell_{\text{P}} = 1.616 \times 10^{-35} \text{ m} \quad (7)$$

Minimal Length Scale

Minimal Length Scale from ξ

$$L_0 = \xi \cdot \ell_{\text{P}} = 2.155 \times 10^{-39} \text{ m} \quad (8)$$

The complete derivation chain $\xi \rightarrow m_e \rightarrow G \rightarrow \ell_{\text{P}} \rightarrow L_0$ depends only on ξ as the single fundamental input. SI numerical values arise through the unavoidable calibration to human measurement units — physically the theory is parameter-free.

.2 The Maximal Scale: Hubble Radius (ξ -notation, calibrated to H_0)

Hubble Energy from ξ

The Hubble energy is written in ξ -notation (Document 026):

$$E_H = E_0 \cdot \xi^{41/4} \quad (9)$$

with $E_0 = \sqrt{m_e m_\mu / K_{\text{frak}}} = 7.398 \text{ MeV}$ and $\xi = 4/30000$. The exponent $41/4$ here is *not* derived from ξ but chosen so as to match the measured H_0 (Doc. 190, R35/R36); $\xi^{41/4}$ is the measured cosmic scale in ξ -notation, not a prediction.

Numerical evaluation:

$$\xi^{41/4} = (1.333 \times 10^{-4})^{41/4} = 1.908 \times 10^{-40} \quad (10)$$

$$E_H = 7.398 \text{ MeV} \times 1.908 \times 10^{-40} = 1.412 \times 10^{-33} \text{ eV} \quad (11)$$

Hubble Constant from ξ

$$H_0^{\text{T0}} = \frac{E_H}{\hbar} = \frac{1.412 \times 10^{-33} \text{ eV}}{6.582 \times 10^{-16} \text{ eV s}} \approx 66.2 \text{ km/s/Mpc} \quad (12)$$

Deviation from the Planck value (67.4 km/s/Mpc): -1.9% .

Hubble Radius as Maximal Scale

Distinction: Hubble radius vs. particle horizon

Standard cosmology (Λ CDM, expanding universe) distinguishes two cosmic horizons: the *Hubble radius* $R_H = c/H_0 \approx 1.4 \times 10^{26} \text{ m}$ and the substantially larger *particle horizon* (observable universe, $\approx 4.4 \times 10^{26} \text{ m} \approx 46.5 \text{ Gly}$), which encompasses the full causal past since the Big Bang. The particle horizon is only meaningfully defined in an expanding universe with a Big Bang.

In T0 theory there is no expansion and no Big Bang. The Hubble radius R_H is the sole relevant geometric horizon: beyond R_H the accumulated fractal path elongation yields $z \rightarrow \infty$, i.e. no information can be received. A particle horizon in the Λ CDM sense does not exist conceptually in T0. Numerical comparisons with standard values must therefore always refer to R_H , not to the particle horizon.

The Hubble radius follows directly from the Hubble energy:

$$R_H = \frac{\hbar c}{E_H} = \frac{1.973 \times 10^{-7} \text{ eV m}}{1.412 \times 10^{-33} \text{ eV}} \quad (13)$$

Hubble Radius from ξ

$$R_H = 1.398 \times 10^{26} \text{ m} \quad (14)$$

In ξ -notation; the exponent $41/4$, however, is calibrated to H_0 , not an independent ξ result (Doc. 190, R35/R36).

Hubble Radius and Total Mass

The total mass of the observable universe corresponding to the Hubble radius:

$$M_U = \frac{R_H \cdot c^2}{2G} = 9.43 \times 10^{52} \text{ kg} = 4.7 \times 10^{22} M_\odot \quad (15)$$

Hubble Radius from ξ

$$R_H = 1.398 \times 10^{26} \text{ m} \quad (16)$$

In ξ -notation; the exponent $41/4$, however, is calibrated to H_0 , not an independent ξ result (Doc. 190, R35/R36).

Scale	Value	Interpretation
$L_0 = \xi \cdot \ell_P$	$2.155 \times 10^{-39} \text{ m}$	Sub-Planck scale
ℓ_P	$1.616 \times 10^{-35} \text{ m}$	Planck length
$\lambda_e = \hbar/m_e c$	$3.86 \times 10^{-13} \text{ m}$	Electron Compton wavelength
$R_H = \hbar c/E_H$	$1.398 \times 10^{26} \text{ m}$	Hubble horizon

Table 1: Scale hierarchy in ξ -notation (cosmological scale R_H calibrated to H_0)

.3 The Complete Scale Hierarchy

Scale Hierarchy

The Ratio of Scales

$$\frac{R_H}{L_0} = \frac{1.398 \times 10^{26} \text{ m}}{2.155 \times 10^{-39} \text{ m}} \quad (17)$$

Scale Ratio from ξ

$$\boxed{\frac{R_H}{L_0} = 6.49 \times 10^{64}} \quad (18)$$

65 orders of magnitude between the minimal and maximal scale — both from $\xi = 4/30000$. (Corrected on 30 Sept. 2026: previously 57 orders of magnitude – $\log_{10}(6.49 \times 10^{64}) = 64.8$.)

The Complete Derivation Chain

Complete Chain from ξ to Both Limits

$$\text{Lagrangian} \Rightarrow \xi = 4/30000$$

$$\Downarrow$$

$$G_{\text{SI}} = \frac{\xi^2}{4m_e} \times C_{\text{conv}} \times K_{\text{frak}} \Rightarrow \ell_P = \sqrt{\hbar G/c^3}$$

$$\Downarrow$$

$$\Downarrow$$

$$L_0 = \xi \cdot \ell_P \quad E_H = E_0 \cdot \xi^{41/4}$$

$$= 2.155 \times 10^{-39} \text{ m} \quad \Downarrow$$

$$R_H = \hbar c/E_H$$

$$= 1.398 \times 10^{26} \text{ m}$$

$$\Downarrow$$

$$\boxed{R_H/L_0 = 6.49 \times 10^{64} \text{ (calibrated to } H_0\text{)}} \quad (19)$$

.4 Physical Meaning

Physical Meaning of R_H

In a static universe without expansion the Hubble radius is the geometric horizon beyond which photons in the ξ field are completely redshifted ($z \rightarrow \infty$) through accumulated fractal path elongation — not through intrinsic energy loss of the photon itself, but through geometric stretching of the propagation path. Tired light does not occur.

Redshift without Expansion and without Frequency Dependence

The redshift arises through fractal summation of the path extension of photons in the T0 vacuum field: $z \approx \xi \cdot \ln(d/\ell_P)$. Since the path extension is a property of the geometry itself — not of the photon — all frequencies experience exactly the same stretching. There is no frequency dependence. The effect is observationally equivalent to an expanding universe but mechanistically fundamentally different. Derived in detail in Document 026 (Geometric Cosmology).

(Note of 30 Sept. 2026: The formula $z \approx \xi \ln(d/\ell_P)$ yields at $d = R_H = 1.4 \times 10^{26} \text{ m}$ only $z = \xi \cdot 140.3 = 0.019$ (with L_0 instead of ℓ_P : 0.020) and grows only logarithmically with d . The statement that $z \rightarrow \infty$ beyond R_H (delimitation box and section on the physical meaning of R_H) is not compatible with this formula.)

Scale relativity of the torus: fractal nesting

In T0 theory the 4D torus is not a cosmic container in which particles reside. Rather, every physical system carries **its own characteristic torus**, whose scale is set by the Compton wavelength $\bar{\lambda} = \hbar/mc$ of that system:

$$R_{\text{torus}}(m) = \frac{\hbar}{mc} \quad (20)$$

System	Characteristic scale	Ratio to L_0
Electron	$3.86 \times 10^{-13} \text{ m}$	$\sim 10^{26}$
Proton	$2.10 \times 10^{-16} \text{ m}$	$\sim 10^{23}$
Hydrogen	$5.29 \times 10^{-11} \text{ m}$	$\sim 10^{28}$
Galaxy	$\sim 10^{21} \text{ m}$	$\sim 10^{60}$
Universe	$R_H = 1.40 \times 10^{26} \text{ m}$	6.49×10^{64}

Table 2: System-specific torus scales — all sharing L_0 as common lower bound

Fractal nesting without rigid hierarchy

These structures are not nested like Russian dolls — the electron torus does not sit *inside* the galactic torus. Each scale is the geometric self-structure of the respective system, constituted by its mass. The common lower bound L_0 is the same at every level; the upper bound is system-specific.

The fractal coupling between scales is mediated by ξ and therefore very small. Interactions between widely separated scales — for instance between electron structure and the cosmological field — are not zero, but suppressed by powers of $\xi \approx 1.33 \times 10^{-4}$. This weak cross-scale coupling is physically relevant: it is the origin

of the fractal correction factors K_{frak} in the calculation of g -factors, particle masses, and G — small but measurable contributions that are absent in the Standard Model or must be introduced as free parameters.

Locality, flat torus, and finiteness of calculations

The T0 torus is not the intuitive donut torus embedded in $\mathbb{R}^3$. It is the **flat identification torus** $T^n = \mathbb{R}^n / \mathbb{Z}^n$: a Euclidean space with opposite boundaries identified. This torus has **intrinsic curvature** $K = 0$ **everywhere pointwise** — not merely in the integral. Locally it is exactly identical to $\mathbb{R}^n$, without any approximation.

Flat torus: locally exactly Euclidean

For the embedded donut torus in $\mathbb{R}^3$, Gaussian curvature varies: positive on the outside ($K > 0$), negative on the inside ($K < 0$). The integral vanishes exactly (Gauss-Bonnet: $\int K dA = 2\pi\chi = 0$ for $\chi = 0$). For the flat identification torus, by contrast, $K = 0$ holds *everywhere* — there is no inner and outer curvature that must cancel in an integral. Calculations within this torus are exactly the same as in unbounded Euclidean space. No curvature corrections, no topological terms.

Three consequences follow for calculations in T0 theory:

1. No renormalisation required. All field integrals run over the finite local torus with scale $R_{\text{torus}}(m) = \hbar/mc$. Since this torus is finite and locally flat, UV divergences are structurally excluded. The upper integration limit is not ∞ but R_{torus} ; the lower limit is L_0 . Both are finite and derived from ξ .

2. The ξ correction for an infinite universe is not negligibly small — it is simply irrelevant. Formally an infinite universe would generate a correction $\sim \xi \cdot \ln(R_\infty/R_H)$, which diverges logarithmically as $R_\infty \rightarrow \infty$. This divergence does not apply, however, because all physical observables are integrals over the *local* torus of the respective system. Contributions from scales $\gg R_{\text{torus}}(m)$ do not couple to local physics — they are suppressed by powers of ξ and lie outside the integration domain. Whether the universe is finite or infinite has no effect on local calculations.

3. Curvature corrections are absolutely negligible. The relative curvature correction at the smallest physical scale L_0 within a torus of scale R_{torus} is $(L_0/R_{\text{torus}})^2$:

$$\begin{aligned} \text{Electron: } \left(\frac{L_0}{\bar{\lambda}_e}\right)^2 &= \left(\frac{2.155 \times 10^{-39}}{3.86 \times 10^{-13}}\right)^2 \approx 3.1 \times 10^{-53} \\ \text{Proton: } \left(\frac{L_0}{\bar{\lambda}_p}\right)^2 &\approx 1.1 \times 10^{-46} \end{aligned}$$

These values are exactly zero for all practical calculations.

One field — masses as knots — emergent coherence length

Behind these three consequences lies a unified ontological structure: there is **one single ξ vacuum field**. There is no separate hierarchy of fields for different scales. Masses are **topological knots** in this field — localised structures, not external objects. Each knot of mass m carries, through the duality $\tilde{T} = 1/m$, a characteristic time scale, and from this its spatial coherence length $\hbar/mc$ emerges immediately. The torus radius is not externally assigned but follows from the knot structure itself.

The hierarchy of torus scales — from the proton knot ($\sim 10^{-16}$ m) to the cosmological horizon ($\sim 10^{26}$ m) — is therefore nothing other than the **fractal self-similarity of a single field**: the same ξ at every scale, the same lower bound L_0 , system-specific coherence lengths as emergent properties of the respective knot structure. The cross-scale coupling is not zero, but suppressed by powers of ξ — which is why one calculates locally without knowing the full field and still obtains exact results.

The exponent $41/4$ in $E_H = E_0 \cdot \xi^{41/4}$ follows from the renormalisation group structure of the T0 field over cosmological scales and still requires full derivation from the ξ field theory (Document 026). The numerical agreement with $H_0 = 66.2$ km/s/Mpc (-1.9% from Planck) is a strong consistency argument.

On the role of R_H in calculations

The precise numerical value of R_H plays a subordinate role in the physical calculations of T0 theory. The structurally significant quantity is $L_0 = \xi \cdot \ell_P$ — it follows purely from ξ and is the true foundation of all scale statements.

R_H , by contrast, is different in two respects. First, its numerical value is tied to the measured H_0 (or depends on the exponent $41/4$ still to be fully derived). Second, R_H in T0 is not primarily a computed result but a **boundary by definition**: the scale beyond which $z \rightarrow \infty$ holds and no observation is possible. More than R_H is structurally inaccessible within the model — regardless of whether the exponent $41/4$ is rigorously derived or fitted to H_0 .

The ratio $R_H/L_0 = 6.49 \times 10^{64}$ therefore does not express a deep theoretical truth, but the distance between the geometrically derived lower bound and the observation-based horizon. The genuine achievement of the theory lies in L_0 — not in R_H .

Cosmological Constant from ξ

The dimensionless cosmological constant follows directly:

$$\Lambda_{\text{dim}} = \left(\frac{\ell_P}{R_H} \right)^2 = \left(\frac{1.616 \times 10^{-35}}{1.398 \times 10^{26}} \right)^2 \approx 1.34 \times 10^{-122} \quad (21)$$

The standard value is $\Lambda_{\text{dim}} \approx 2.87 \times 10^{-122}$ (from $\Lambda \approx 1.1 \times 10^{-52} \text{ m}^{-2}$, CODATA). The T0 result is in the same order of magnitude (factor 2.1) — not dark energy, but a geometric consequence of the scale hierarchy from ξ . Full agreement requires the still-pending derivation of the exponent $41/4$ (Document 026).

.5 Summary

Complete Scale Hierarchy from $\xi = 4/30000$

Lower limit:	$L_0 = \xi \cdot \ell_P = 2.155 \times 10^{-39} \text{ m}$
Planck:	$\ell_P = 1.616 \times 10^{-35} \text{ m}$
Electron:	$\lambda_e = \hbar/m_e c = 3.86 \times 10^{-13} \text{ m}$
Hubble:	$R_H = \hbar c/E_H = 1.398 \times 10^{26} \text{ m}$

All scales from $\xi = 4/30000$ —no free parameter

 (22)

The universe is geometrically self-consistently closed — not by a boundary, but by the topology of the 4D torus with L_0 and R_H as natural limits of the same geometry.

Key conclusion: Is there a maximum size of the universe?

The question is not meaningfully posed as a global question in FFGFT. Every physical scale — from the proton to the cosmological horizon — possesses its **own system-specific upper bound** $R_{\text{torus}}(m) = \hbar/mc$, emergent from the knot structure of the respective mass in the single ξ field. R_H is the upper bound of the cosmological sector. The universal lower bound L_0 applies equally to all scales — a universal upper bound does not exist, because there is no scale-independent physics for which it would be relevant.

Epistemological framing: model boundaries, not existence boundaries

L_0 and R_H are not physical walls and make no claim about the total size or global structure of the universe. They are **boundaries of describability** within T0 theory: below L_0 the spacetime granulation diverges, beyond R_H the redshift diverges ($z \rightarrow \infty$). Whether anything exists outside these boundaries is simply not a question the theory addresses.

The decisive advantage over other approaches lies not in this epistemological modesty alone, but in the fact that T0 **requires no regularisation tricks whatsoever**: no renormalisation procedures to remove UV divergences, no artificial cutoffs, no counterterms, no auxiliary fields introduced to tame infinities. L_0 does not arise as a post-hoc correction — it emerges as a direct geometric consequence of ξ . The theory is finite from the outset, not because infinities have been subtracted away, but because the parameter ξ structurally excludes them.

Note: Connection to the SI System

The absolute numerical values in this document — L_0 , R_H , M_U in SI units — presuppose a connection to the SI unit system. This is not circular but epistemologically necessary: from a dimensionless ξ alone, no absolute metre or kilogram values can be derived. The connection is made via m_e , ℓ_P or α as equivalent entry points. Treated in detail in Document 180 (Section: Connection to the SI System) and in Documents 056 and 101. Documents 180, 181 and 182 form a unit and should not be read separately.

The exponent $41/4$ still requires full derivation from the ξ field theory (Document 026).

Addendum P35: ξ^N Triviality (7 September 2026)

General clarification (P35, corpus-wide): Expressions of the form " $X = \xi^N$ " are *intrinsically meaningless on their own*, since any positive quantity can be written as a ξ -power — an exponent alone is not a prediction.

What makes a statement non-trivial: (i) The *form* is derived forward (geometrically or algebraically forced), not fitted after the fact. (ii) The *exponent* is rational and readable from

the structure (e.g. $41/4$ from T^4 geometry), not chosen freely. (iii) The number (prefactor, anchor) is an *external calibration* and declared as such (cf. P20, P33, P39).

For this document: The exponents of the scale hierarchy ($R_H \sim \xi^{-41/4} \ell_P$ etc.) are geometrically plausible **[B]**, but the cosmic exponent $41/4$ has not yet been derived forward (P20). Statements of this kind are to be classified as **[S]**, not **[K]**.